

Clinic Application Data Entry Requirements Report

Complete current data-point catalog, validation rules, and recommended missing clinical/operational fields

Audit item	Value
Application	Clinic Management System (Flutter web + Flask/MySQL API)
Production URL	https://clinic.nalexustechologies.com
Audit date	16 June 2026 (Asia/Karachi)
Scope	Local repository, deployed server, live public endpoint, database state, and static code review
Security note	Passwords, API keys, database credentials, and secret values are intentionally excluded.

Purpose

This report lists the data that users, administrators, staff, and integrations should enter into the Clinic Management System. It separates fields currently supported by the application from important data points that should be added before the system is used as a complete clinical record.

Current Application Data Points

Area / record	Data points to enter	Required / validation / notes
Login	Email; password	Both required; valid active user; clinic must be approved for non-super-admin roles
Password change	Current/old password; new password; confirmation (UI should add)	New password >= 6 chars and different; frontend/backend key mismatch must be fixed
Clinic core	Clinic name; owner name; admin email; phone; address; city; logo	Name, owner, email, phone required; email globally unique; logo exists in model but has no working UI
Clinic operations	Clinic type; number of doctors; has pharmacy; has receptionist; opening time; closing time; working days	Type single_doctor/multi_doctor; doctor count must match; times HH:MM; closing after opening should be enforced
Clinic status/subscription	Status; subscription plan; approval actor/date; subscription start/end; amount paid	Admin-controlled; plan assignment and subscription lifecycle UI/API are missing
Doctor account/profile	Name; email; phone; department; specialization; qualification; experience years; license number; consultation fee; available days; start/end time; status	Name/email required; email unique; department selected/created; non-negative fee/experience recommended
Receptionist user	Name; email; phone; status	Name/email required; valid unique email; temp password generated
Pharmacy user	Name; email; phone; status	Name/email required; valid unique email; clinic pharmacy must be enabled
Assistant	Assigned doctor; name; email; phone; duties; six permission flags; status	Doctor/name/email required; email unique; permissions control workflow endpoints
Patient	Name; age; gender; phone; CNIC; address; blood group; emergency contact; linked user ID	Backend requires name and phone; phone/CNIC unique per clinic; gender male/female/other
Appointment	Patient; doctor; date; time; consultation type; fee; payment status; paid amount; payment method; notes	Patient/doctor/date/time required; type new/followup/emergency; paid/partial rules enforced
Queue/status	Appointment status; cancel reason; reschedule date/time	Role-restricted transitions; status waiting/sent_to_assistant/in_consultation/completed/cancelled
Vitals	Patient; appointment; temperature C; blood pressure; pulse; weight kg; height cm; oxygen %; notes	Patient required; appointment optional but must match doctor/patient; numeric validation is limited
Patient report	Patient; appointment; report title; report type; file URL; notes	Patient and title required; current implementation stores URL, not uploaded file
Consultation draft	Patient; appointment; symptoms draft; vitals summary; notes	Assistant permission-controlled; draft supports doctor workflow
Prescription	Appointment; patient; symptoms; diagnosis; notes; follow-up date; medicines; lab tests	Appointment/patient required; one prescription per appointment; doctor ownership enforced
Prescription medicine	Inventory medicine ID; medicine name; dosage; frequency; duration; instructions	Medicine name required; inventory ID optional
Prescription lab test	Test name; instructions	Test name required
Inventory medicine	Medicine name; category; batch number; expiry date; purchase price; sale price; quantity; supplier; rack; low-stock limit; status	Name, sale price, quantity required by backend; duplicate name+batch blocked; values non-negative
Pharmacy sale	Patient; prescription; sale items; payment status; payment method	Patient/prescription optional; item medicine ID and quantity required; stock must be available
Pharmacy sale item	Medicine ID; quantity; unit price/total derived	Quantity positive; price and total should be server-derived

Area / record	Data points to enter	Required / validation / notes
General payment	Patient; appointment; payment type; amount; method; status	Type consultation/pharmacy/lab/other; amount > 0; method allowed; status paid/pending/refunded
Reports/filters	Start date; end date; group by; clinic/doctor/status/type/method filters; export flag	Read-only input; validate date range and authorized scope
Clinical AI input	Patient ID; appointment ID; symptoms; diagnosis; notes; vitals summary; pasted medical text	Doctor-only; patient/appointment required by task; clinician must review all outputs

Detailed Allowed Values and Rules

Field group	Allowed values / format	Rule
User role	super_admin, clinic_admin, doctor, assistant, receptionist, pharmacy, patient	Role controls navigation and API authorization
User status	active, inactive, pending	Inactive/pending users cannot use normal flows
Clinic type	single_doctor, multi_doctor	Single doctor must provide exactly one doctor
Clinic status	pending, approved, suspended	Non-super-admin login requires approved clinic
Department status	active, inactive	Name unique within clinic
Doctor/assistant status	active, inactive	Inactive staff should not be selectable
Gender	male, female, other	Optional in current backend
Blood group	A+, A-, B+, B-, AB+, AB-, O+, O-	Validated by UI and backend
Date	YYYY-MM-DD	Backend parser format
Time	HH:MM, 24-hour	Backend parser format
Appointment type	new, followup, emergency	Defaults to new
Appointment status	waiting, sent_to_assistant, in_consultation, completed, cancelled	Transitions restricted by role
Appointment payment status	unpaid, paid, partial	Paid amount/method rules apply during booking
Payment type	consultation, pharmacy, lab, other	Role-restricted when creating payments
Payment method	cash, card, easypaisa, jazzcash, bank	Frontend/backend field name must be method for general payment
General payment status	paid, pending, refunded	Partial appointment balances are represented by multiple paid ledger entries and appointment payment_status
Prescription pharmacy status	pending, processing, dispensed, cancelled	Pharmacy order workflow
Medicine status	active, inactive	Inactive items excluded from normal use
Sale payment status	paid, pending	Confirm against pharmacy service before rollout
Email	Standard email format	Globally unique for user accounts
Phone/CNIC	Text	Patient phone and CNIC are unique within a clinic; stricter format validation should be added
AI provider/model	openai / gpt-5.2 currently	Server-side only; never enter provider API keys in the browser
Disease risk model	diabetes, heart_disease, stroke, breast_cancer	Doctor-only educational scikit-learn demo endpoints

Recommended Missing Patient and Clinical Data

The current patient model is too small for a complete longitudinal medical record. The following should be added with clear ownership, consent, retention, and access controls.

Category	Recommended data points	Priority / reason
Patient identity	Date of birth instead of age; preferred name; photo; marital status; occupation; language; nationality	High: age becomes inaccurate; identity and communication needs
Contact	Email; multiple phones; preferred contact method; guardian/caregiver; emergency contact name/relationship	High: current emergency contact stores phone only
Address	Structured street, area, city, province, postal code, country	Medium: improves billing and reporting
Allergies	Substance; reaction; severity; status; date recorded; recorder	Critical: required for safer prescribing
Problem list	Condition; ICD/SNOMED code; onset; status; severity; notes	Critical: longitudinal diagnosis tracking
Medical history	Past illnesses; surgeries; hospitalizations; family history; social history	High: needed for clinical decisions
Current medication	Medicine; dose; route; frequency; indication; start/end; adherence	Critical: prescriptions alone do not represent current medicines
Immunizations	Vaccine; dose; date; manufacturer/lot; site; next due	High for general practice
Pregnancy/reproductive	Pregnancy status; LMP; estimated due date; breastfeeding	Critical where clinically applicable
Clinical observations	Respiratory rate; pain score; BMI; glucose; consciousness; waist; units and reference ranges	High: current vitals are incomplete
Encounter	Chief complaint; history of present illness; review of systems; examination; assessment; plan; disposition	Critical: current prescription notes are not a full encounter note
Diagnosis coding	Primary/secondary diagnosis; code; certainty; present-on-admission; resolved status	High: reporting/interoperability
Orders/results	Lab/imaging order; specimen; result values; units; reference range; abnormal flag; status; performer; result date	Critical: current lab tests are text-only orders
Documents	Actual file upload; MIME type; checksum; size; category; capture date; author; version	High: current patient report stores only URL/title
Referral	Referred-to provider/facility; reason; urgency; status; appointment/result	Medium
Consent/privacy	Consent type; scope; signed date; expiry; withdrawal; privacy notice acknowledgment	Critical for clinical and AI processing
Clinical communication	Patient instructions; education; language; acknowledgment; secure messages	Medium

Recommended Missing Operational and Financial Data

Area	Recommended data points	Reason
Clinic settings	Timezone; currency; tax ID; registration/license; invoice prefix; receipt footer; cancellation policy; holidays; branding	Basic clinic identity/hours/days settings are implemented; these expanded fields remain recommended
Staff	Address; national ID; designation; employment dates; shift; payroll/rate; credentials; credential expiry; emergency contact	Required for complete staff administration
Doctor scheduling	Slot duration; breaks; leave; holidays; capacity; appointment modes; location/room	Current availability is only days and start/end time
Appointments	Booking source; reason for visit; priority; check-in/out; no-show; cancellation actor/time; reminder status	Needed for operations and analytics
Payments	Invoice number; billed amount; discount; tax; balance; payer; transaction/reference ID; refund reason; notes; reconciliation status	Current Payment ledger is minimal; billing contract and appointment balance synchronization are implemented
Insurance	Payer; member/policy ID; coverage; authorization; claim; status; approved/denied amount	Absent
Pharmacy purchasing	Supplier contact; purchase order; goods receipt; purchase invoice; tax; discount; batch received quantity; return	No procurement workflow
Inventory safety	Manufacturer; generic/brand; strength; form; barcode; reorder point/quantity; controlled-drug flag; storage condition	Current inventory fields are basic
Subscription	Assigned plan; billing cycle; start/end; renewal; status; amount; payment reference; limits/usage	Plan CRUD and assignment exist; renewal/payment automation and usage metering remain recommended
Audit/security	Actor; action; entity type/id; before/after values; IP; user agent; timestamp; correlation ID; AI provider/model/prompt version	Baseline actor/action/module/IP logging exists; entity diffs, correlation IDs, and AI-specific traceability remain recommended
Notifications	Recipient; channel; template; scheduled/sent time; delivery/read status; failure reason	No reminder/notification tracking

Clinical AI Data Requirements and Governance

Data point	Current state	What should be entered/stored
AI request context	Patient history, appointment, working note, vitals summary	Only minimum necessary data; avoid unnecessary identifiers
AI provider/model	OpenAI / gpt-5.2 configured	Provider, exact model/version, request timestamp, feature used
scikit-learn model	Four educational demos implemented	Model key, algorithm, feature schema, demo accuracy, risk probability, and safety notice
Prompt/version	Embedded in code only	Prompt template ID/version and safety-policy version
Output	Returned to UI; not auto-saved	Store only after clinician approval, clearly marked as AI-assisted
Approval	Doctor applies draft manually	Approver user, approval time, edits/diff, final disposition
Failure/fallback	Rule-based fallback and provider_error returned	Provider latency, status, fallback reason; never log raw PHI carelessly
Consent/legal basis	Not represented	Clinic/provider agreement, patient consent/legal basis, retention rule
Usage/cost	Not represented	Token/usage estimate and per-clinic cost/limits where permitted

Data Quality and Validation Corrections

- Patient phone is visibly required in the UI and backend; add jurisdiction-specific normalization if required.
- Use date of birth and calculate age; do not store age as the primary demographic value.
- Validate phone, CNIC, blood group, vital ranges, dosage, duration, and clinical units consistently.
- Change-password payload and forced first-login password workflow are aligned and implemented.
- Standalone billing payload, payment model response mapping, and appointment balance synchronization are aligned.
- Relational checks exist for appointment patient/doctor/clinic, report appointment/patient, and sale prescription/patient.
- Server-derived pharmacy prices and payment balance updates are implemented; add reconciliation controls as finance scope grows.
- Baseline mutation audit history is implemented; add before/after values and explicit record ownership fields where needed.
- Use actual secure file upload/storage rather than arbitrary report URLs.
- Define data retention, deletion, backup, restoration, and cross-clinic isolation policies.

Minimum Dataset Before Go-Live Testing

Dataset	Minimum test entries
Organization	1 approved clinic with settings and an assigned subscription plan
Users	1 account for each role; all temporary passwords changed
Doctors/departments	At least 2 doctors in 2 departments with availability
Patients	At least 5 synthetic patients covering demographics, allergies, histories, and contact variations
Appointments	New, follow-up, emergency, paid, partial, cancelled, rescheduled, and completed examples
Clinical	Vitals, reports, prescriptions with medicines/lab tests, follow-ups, and doctor notes
Pharmacy	Active, low-stock, expiring, expired, duplicate-batch, prescription, and walk-in sale examples
Finance	Paid, pending, partial/booked, refunded, consultation, pharmacy, lab, and other payments
AI	Synthetic patient summary, consultation draft, provider failure/fallback, red-flag, and clinician-rejection examples
Security	Cross-clinic denial, role denial, inactive account, suspended clinic, expired token, and audit-log examples

Source of Truth

Current fields and rules were derived from the SQLAlchemy models, Flask services/routes/validators, Flutter forms/services, schema.sql, and the deployed production database. Recommended missing fields are requirements for a more complete operational and clinical system; they are not currently implemented.